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Convergence of the Least Squares Monte Carlo Approach to American Option Valuation

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In a recent paper, Longstaff and Schwartz (2001) suggest a method to American option valuation based on simulation. The method is termed the Least Squares Monte Carlo (LSM) method, and although it has become widely used, not much is known about the properties of the estimator. This paper corrects this shortcoming using theory from the literature on semiparametric series estimators. A central part of the LSM method is the approximation of a set of conditional expectation functions. We show that the approximations converge to the true expectation functions under general assumptions in a multiperiod, multidimensional setting. We obtain convergence rates in the two-period, multidimensional case, and we discuss the relation between the optimal rate of convergence and the properties of the conditional expectation. Furthermore, we show that the actual price estimates converge to the true price. This provides the mathematical foundation for the use of the LSM method in derivatives research.

Key words: American options; Monte Carlo methods; least squares regression

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1. Introduction

It is well known that the price of an option generally depends on the strike price, the value of the underlying asset, the volatility of this asset, the amount of dividends paid on it, the interest rate, and the time to maturity (see Hull 1997 for a thorough discussion of the effect of each of these factors). To price traded options correctly, it is thus necessary to take all these factors into account. Furthermore, when the option is American, the possibility of early exercise should be considered and an optimal early exercise policy must be determined. This often leads to highly complicated calculations, and it is seldom possible to find analytical solutions to the formulae.

When analytical solutions cannot be found, one has to use numerical methods. The most famous of the numerical methods is, without doubt, the binomial model suggested by Cox et al. (1979). Although it is possible to incorporate the early exercise features in the binomial model, it is not computationally feasible to handle more than a couple of stochastic factors. The problem is that the number of nodes required grows exponentially if we wish to allow for multiple stochastic factors such as interest rates, dividends, volatilities, or multiple underlying assets. Other methods like the finite difference method simply cannot be extended to more than two or three stochastic factors. An alternative suggestion is to use simulation techniques. Simulation techniques have been used to

price options for quite some time (see Boyle 1977), but in most cases, the options have been European and not American. Actually, it has been the general idea until recently that it would be impossible from a computational point of view to use simulation methods to price American options (see Campbell et al. 1996, Hull 1997). The reason is that the optimal early exercise policy must be calculated recursively. Using simulation techniques, at any time along any of the paths, there is only one future path, thus, rendering the determination of an optimal early exercise strategy difficult.

One of the first to propose solutions to the problem of pricing American options using simulation, and in particular of determining the optimal early exercise strategy, was Tilley (1993). In that paper, a simulation algorithm that mimics the standard lattice method of determining the value of holding the option as the present value of the expected one-period-ahead option value is presented. Barraquand and Martineau (1995) develop a method that is closely related to that of Tilley (1993), but easier to extend. The idea is to partition the state space of simulated paths into a number of cells in such a way that the payoff from the option is approximately equal across the paths in the particular cell. The probabilities of moving to different cells next period conditional on the current cell can then be calculated from the simulated paths. With these probabilities, the expected value of keeping the

option alive until next period can be calculated, and a strategy for optimal exercise determined.¹

An alternative way to formulate the American option pricing problem is in terms of optimal stopping times. This is done in Carriere (1996), where it is shown, by use of a backwards induction theorem, that pricing an American option is equivalent to calculating a number of conditional expectations. These are generally difficult to compute, but the paper shows how to combine simulation methods with advanced regression methods to get an approximation. A new and somewhat simpler simulation-based method to price American options has recently been proposed by Longstaff and Schwartz (2001). The idea is to estimate the conditional expectation of the payoff from continuing to keep the option alive at each possible exercise point from a cross-sectional least squares regression using the information in the simulated paths. The paper shows how to price different types of path-dependent options using this method. The method is rapidly gaining importance in a variety of applied areas, e.g., real option valuation (see Gamba 2002). Although the performance of the method has been examined in some detail (see Moreno and Navas 2003), little is known about the asymptotic behavior of the LSM estimator. The paper by Longstaff and Schwartz (2001) provides us with a convergence result only in the simplest possible situation with one state variable and one possible early exercise time. The recent paper by Clément et al. (2002) proves convergence of the algorithm to an approximation to the true price. The reason that approximation is involved, even in the limit, is that the relevant cross-sectional least squares regression only uses a fixed and finite number of regressors. A somewhat related method is proposed in Tsitsiklis and Van Roy (2001), in which a recursive algorithm of the same type is proposed. The main difference to Longstaff and Schwartz (2001) is in terms of what is approximated, because Tsitsiklis and Van Roy approximate the value function directly and not the conditional expectations. Again, Tsitsiklis and Van Roy only consider a fixed and finite number of regressors. Hence, the method does not produce the true value even in the limit.

In this paper, we generalize the convergence results of Longstaff and Schwartz (2001) and Clément et al. (2002) and prove convergence of the LSM algorithm to the true price. The key to our results is to allow the number of regressors in the cross-sectional least squares regression to tend to infinity, along with the number of simulated paths, in carefully chosen proportions. Our focus is on the conditional expectation

approximation. We show that with the convergence of the conditional approximation function, it follows that the price estimate from the LSM method converges to the true price. Our first theorem proves the convergence of the conditional expectation approximation to the true expectation in a mean square sense in the two-period case using results from the literature on semiparametric series estimators. This theorem holds more generally than the one in Longstaff and Schwartz (2001) and, in particular, it allows for multiple state variables. Next, we extend this result to the multiperiod setting, and prove convergence of the approximation for more than one early exercise point. With this theorem, we have provided the mathematical foundation for the use of the LSM method in derivatives research. Furthermore, our first theorem provides us with rates of convergence in the situation with one early exercise time. This yields useful insights into the relation between the optimal rate of convergence and the properties of the conditional expectation.

The structure of this paper is as follows. Section 2 describes how to price options using simulation techniques, and introduces the LSM method. Section 3 contains the main theorems and discussion of the results. Section 4 concludes. Proofs may be found in the appendix.

2. Simulation and Option Pricing

A major problem with the existing numerical methods to option valuation is that they are not easily extended to more than a couple of stochastic factors. In the binomial model, the reason is that in practice, the number of nodes required grows exponentially in the number of stochastic factors. A possible solution to this "curse of dimensionality" is to use simulation. To see this, consider an option written on not one, but r assets. In the binomial model, there will be 2^r branches emanating from each knot in the tree, and the total number of knots grows exponentially in r . However, when using a simulation approach, new values of the r stocks are simply random draws from some prespecified distribution and, as such, the number of nodes remains constant through time and the total number grows only linearly with r . Thus, even with, e.g., $N = 100,000$ paths in the simulation, the number of nodes and, therefore, the number of calculations is not necessarily computationally prohibitive. The same holds if we want to allow for stochastic interest rates, volatility, or dividends.

Simulation methods are immediately applicable to the pricing of European options, and the method was actually introduced even before the binomial model (see Boyle 1977). It is well known that the estimated price is unbiased and asymptotically normal, and the

¹ Broadie and Glasserman (1997) also use simulation to price American options, but their approach is more closely related to the binomial model.

valuation procedure is easily extended to more complex types of European options, such as options on multiple stocks or options whose payoff depends on a function of the path of the underlying asset and not just the terminal value.

Unfortunately, things are not quite as simple when American options are considered. The problem is the need to simultaneously determine the optimal exercise strategy. We may write the price of an American put option with strike price $\bar{S}$ as

$$V(0) = \max_{\tau \leq T} E[e^{-r\tau} \max(\bar{S} - S(\omega, \tau), 0)], \quad (1)$$

where the maximization is over stopping times $\tau \leq T$ adapted to the filtration generated by the relevant stock price process $S(\omega, t)$, and r is the discount rate (typically the riskless rate of interest), which we assume deterministic and constant for simplicity. The problem when trying to estimate this value is that at any possible exercise time, the holder of an American option should compare the payoff from immediate exercise to the expected payoff from continuation. The optimal decision is to exercise if the value of immediate exercise is positive and larger than or equal to the expected payoff from continuation. However, simply using next period's value of the underlying asset to determine the pathwise expected value of keeping the option alive would lead to biased price estimates, as this corresponds to assuming that the holder of the option has perfect foresight (see Broadie and Glasserman 1997). Instead, Longstaff and Schwartz (2001) suggest to estimate the conditional expectation of the payoff from continuing to keep the option alive using the cross-sectional information in the simulation. Below, we formulate the pricing problem in discrete time and present the solution in terms of optimal stopping times.

2.1. Discrete Time Valuation Framework

The first step in implementing any numerical algorithm to price American options is to assume that time can be discretized. Thus, we will assume that the derivative expires in K periods, and specify the early exercise points as $t_0 = 0 < t_1 \leq t_2 \leq \dots \leq t_K = T$. We assume a complete probability space $(\Omega, \mathcal{F}, P)$ equipped with a discrete filtration $(\mathcal{F}(t_k))_{k=0}^K$. The underlying model is assumed to be Markovian, with state variables $(X(\omega, t_k))_{k=0}^K$ adapted to $(\mathcal{F}(t_k))_{k=0}^K$. We denote by $(Z(\omega, t_k))_{k=0}^K$ an adapted payoff process for the derivative, satisfying $Z(\omega, t_k) = h(X(\omega, t_k), t_k)$, for a suitable function $h(\cdot, \cdot)$. As an example, consider the American put option from above, for which the only state variable of interest is the stock price, $X(\omega, t_k) = S(\omega, t_k)$. We have that $Z(\omega, t_k) = \max(\bar{S} - S(\omega, t_k), 0)$. We assume that $X(\omega, 0) = x$ is known and, hence, $Z(\omega, 0)$ is deterministic.

In the literature, it has become standard to limit attention to square-integrable payoff functions. That is, we consider payoff functions for which

$$E[Z^2] = \int_{\Omega} Z(\omega, \cdot)^2 dP(\omega) < \infty.$$

The space of square integrable functions is often denoted $L^2(\Omega, \mathcal{F}, P)$, and it is an example of a Hilbert space, i.e., a complete space equipped with an inner product norm. From the payoff function we can define the function $C(\omega, \tilde{\tau}(t_k)) = e^{-r(\tilde{\tau}(t_k) - t_k)} Z(\omega, \tilde{\tau}(t_k))$ as the cash flow generated by the option, discounted back to t_k and conditional on following a stopping strategy from t_k to expiration, written as $\tilde{\tau}(t_k)$ (essentially this corresponds to the $C(\omega, s; t, T)$ function from Longstaff and Schwartz (2001) defined in terms of stopping times and appropriately discounted). With this formulation, we can specify the object of interest as a slight generalization of (1) and write it as

$$V(0) = \max_{\tilde{\tau}(0) \in \mathcal{T}(0)} E[C(\omega, \tilde{\tau}(0))], \quad (2)$$

where the maximization is over stopping times $\tilde{\tau}(0) \in \mathcal{T}(0)$, with $\mathcal{T}(t_k)$ denoting the set of all stopping times with values in $\{t_k, \dots, t_K\}$.

2.1.1. Valuation as an Optimal Stopping Time Problem. Problems like (2) are referred to as discrete time optimal stopping time problems, and the preferred way to solve them is to use the dynamic programming principle. For the American option pricing problem, this can be written in terms of the optimal stopping times $\tau(t_k)$ as follows:

$$\begin{cases} \tau(t_K) = T, \\ \tau(t_k) = t_k 1_{\{Z(\omega, t_k) \geq E[e^{-r(t_{k+1} - t_k)} C(\omega, \tau(t_{k+1})) | X(\omega, t_k)]\}} \\ \quad + \tau(t_{k+1}) 1_{\{Z(\omega, t_k) < E[e^{-r(t_{k+1} - t_k)} C(\omega, \tau(t_{k+1})) | X(\omega, t_k)]\}} \end{cases} \quad k \leq K-1. \quad (3)$$

This notation highlights the fact that if we know how to determine the conditional expectations given by $E[e^{-r(t_{k+1} - t_k)} C(\omega, \tau(t_{k+1})) | X(\omega, t_k)]$, we can value the option.² The value of the option in (2) can be expressed in terms of the optimal stopping times in (3) as

$$V(0) = E[C(\omega, \tau(0))]. \quad (4)$$

As a special case, we have the European option, which has an optimal stopping time given by $\tau(0) = T$. With this, the price of the option would be

$$V(0) = p(0) = E[e^{-rT} Z(\omega, T)].$$

² In the paper by Tsitsiklis and Van Roy (2001), an alternative specification of the dynamic programming principle using the value functions directly is used. With these, the algorithm generates the value functions iteratively according to $V(T) = Z(\omega, T)$ and $V(t_k) = \max(Z(\omega, t_k), e^{-r(t_{k+1} - t_k)} E[V(t_{k+1}) | X(\omega, t_k)])$, $k \leq K-1$.

The problem with the above formula is that we do not know the conditional expectation function. However, the theory on Hilbert spaces tells us that any function belonging to this space can be represented as a countable linear combination of basis vectors for the space (see Royden 1988). In particular, this is the case for the conditional expectation of a variable belonging to a Hilbert space (like the payoff function). Following Longstaff and Schwartz (2001), we write $F(\omega, t_k) = E[e^{-r(t_{k+1}-t_k)}C(\omega, \tau(t_{k+1}))|X(\omega, t_k)]$, and we have

$$F(\omega, t_k) = \sum_{m=0}^{\infty} \phi_m(X(\omega, t_k))a_m(t_k), \quad (5)$$

where $\{\phi_m(\cdot)\}_{m=0}^{\infty}$ form a basis. Let $F_M(\omega, t_k)$ denote the approximation to $F(\omega, t_k)$ using the first M terms. That is,

$$F_M(\omega, t_k) = \sum_{m=0}^{M-1} \phi_m(X(\omega, t_k))a_m(t_k). \quad (6)$$

The optimal stopping time derived using this approximation, denoted τ_M , can be written as

$$\begin{cases} \tau_M(t_K) = T, \\ \tau_M(t_k) = t_k 1_{\{Z(\omega, t_k) \geq F_M(\omega, t_k)\}} \\ \quad + \tau_M(t_{k+1}) 1_{\{Z(\omega, t_k) < F_M(\omega, t_k)\}}, \quad k \leq K-1, \end{cases} \quad (7)$$

and an approximation to the option value can be calculated as

$$V_M(0) = E[C(\omega, \tau_M(0))]. \quad (8)$$

However, the coefficients $\{a_m(t_k)\}_{m=0}^{M-1}$ in (6) are generally not known and must be estimated when it comes to actually implementing the procedure. Consider the procedure of estimating these using $N \geq M$ simulated paths, each of which we denote $X(\omega_n) = \{X(\omega_n, t_k)\}_{k=0}^K$, $1 \leq n \leq N$, conditionally on $X(\omega_n, 0) = x$. Furthermore, denote the coefficients $\hat{a}_M^N(t_k, \cdot) = \{\hat{a}_m^N(t_k, \cdot)\}_{m=0}^{M-1}$, $0 \leq k \leq K$, where “ $\cdot$ ” is determined by the estimation procedure below. From these, we construct an approximation, $\hat{F}_M^N(\omega, t_k)$, to $F_M(\omega, t_k)$, defined by

$$\hat{F}_M^N(\omega, t_k) = \sum_{m=0}^{M-1} \phi_m(X(\omega, t_k))\hat{a}_m^N(t_k, \cdot), \quad (9)$$

and like before, we can derive an estimate of the optimal stopping time in (7) as

$$\begin{cases} \hat{\tau}_M^N(t_K) = T, \\ \hat{\tau}_M^N(t_k) = t_k 1_{\{Z(\omega, t_k) \geq \hat{F}_M^N(\omega, t_k)\}} \\ \quad + \hat{\tau}_M^N(t_{k+1}) 1_{\{Z(\omega, t_k) < \hat{F}_M^N(\omega, t_k)\}}, \quad k \leq K-1. \end{cases} \quad (10)$$

From (10), a natural approximation to the option value can be calculated as

$$V_M^N(0) = E[C(\omega, \hat{\tau}_M^N(0))]. \quad (11)$$

2.1.2. The Least Squares Monte Carlo Method.

The particular approximation method considered by Longstaff and Schwartz (2001) is that of least squares regression, hence, the terminology LSM method. The algorithm is recursive, and at each point in time t_k , the coefficients $\{\hat{a}_m^N(t_k, \cdot)\}_{m=0}^{M-1}$ are calculated as the solution to the following minimization problem:

$$\min_{\{\hat{a}_m^N\}_{m=0}^{M-1}} \sum_{n=1}^N [\hat{a}_0^N \phi_0(X(\omega_n, t_k)) + \cdots + \hat{a}_{M-1}^N \phi_{M-1}(X(\omega_n, t_k)) - e^{-r(t_{k+1}-t_k)}C(\omega_n, \hat{\tau}_M^N(t_{k+1}))]^2. \quad (12)$$

Using these values in (9) yields the best linear approximation to $F_M(\omega, t_k)$ based on the N given paths.³ In particular, the approximation in (11) reduces to

$$V_M^N(0) = \frac{1}{N} \sum_{n=1}^N C(\omega_n, \hat{\tau}_M^N(0)), \quad (13)$$

because in the last regression ($t_k = 0$), we have $X(\omega_n, t_k) = x$ for all n .

Note that, because $\hat{\tau}_M^N(t_{k+1})$ is determined in (10) using the values of $\hat{a}_M^N(t_{k+1}, \cdot)$, the optimal values in (12) are functions of these and we may write $\hat{a}_M^N(t_k, \hat{a}_M^N(t_{k+1}))$. Furthermore, note that the discounted payoff along a path from time t_k until expiration from following an estimated optimal stopping time, $C(\omega_n, \hat{\tau}_M^N(t_k))$, can equivalently be written as $C(\omega_n, \hat{a}_M^N(t_k, \hat{a}_M^N(t_{k+1})))$, as the stopping time is determined from these coefficients. In particular, this highlights the fact that the coefficients alone determine the approximate value in (13). Realizing the recursive nature of the pricing problem, in the following, we will denote the set of estimated coefficients from time t_k through expiration by $\hat{a}_M^N(t_k) = \{\hat{a}_M^N(t_j, \hat{a}_M^N(t_{j+1}))\}_{j=k}^K$. We will use the same notation for the set of coefficients from (6) $\{a_m(t_k)\}_{m=0}^{M-1}$, $0 \leq k \leq K$, such that $a_M(t_k) = \{a_m(t_j)\}_{j=k}^K$, and for the set of coefficients from (5) $\{a_m(t_k)\}_{m=0}^{\infty}$, $0 \leq k \leq K$, such that $a(t_k) = \{a(t_j)\}_{j=k}^{\infty}$. Also, we let $C(\omega, a_M(t_k))$ and $C(\omega, a(t_k))$ denote the payoff from using the stopping time derived using the respective coefficients.

3. Convergence of the LSM Algorithm

The ultimate goal of this paper is to show that the price estimate from the LSM method described in the previous section converges to the true price. It seems

³ In the paper by Tsitsiklis and Van Roy (2001), what is approximated is the value functions. Hence, in the regressions $e^{-r(t_{k+1}-t_k)} \max(Z(\omega_n, t_{k+1}), \hat{a}_0^N(t_{k+1}, \cdot) \phi_0(X(\omega_n, t_{k+1})) + \cdots + \hat{a}_{M-1}^N(t_{k+1}, \cdot) \phi_{M-1}(X(\omega_n, t_{k+1})))$ is used instead of $e^{-r(t_{k+1}-t_k)}C(\omega_n, \hat{\tau}_M^N(t_{k+1}))$. The authors (like Longstaff and Schwartz 2001) only consider a fixed number of regressors M , which is what we generalize below to achieve convergence.

plausible that the price estimate in (13) will converge to the true price when the conditional expectation approximation converges to the true expectation. However, to show this, we have to ensure that the optimal early exercise strategy determined from the approximation is the correct one. The following assumption states conditions under which this can be ensured.

ASSUMPTION 1. (i) *The simulated paths, $X(\omega_n)$, $1 \leq n \leq N$, are independent, and* (ii) *$\Pr(Z(\omega, t_k) = F(\omega, t_k)) = 0$, $0 \leq k \leq K$.*

The first part of Assumption 1 is, to a certain extent, exactly what a Monte Carlo simulation is: a set of independent random paths. The second part of Assumption 1, on the other hand, is more subtle. This part of the assumption corresponds to what can be found in Clément et al. (2002), but no intuitive explanation was offered in that paper. To get some intuition as to what this assumption means, consider a situation where, for some $\bar{\omega}$, $F(\bar{\omega}, t) = Z(\bar{\omega}, t)$, and although $F_M^N(\omega, t)$ converges to $F(\omega, t)$, it is always the case that $F_M^N(\bar{\omega}, t) > F(\bar{\omega}, t)$. Thus, it is suboptimal to exercise at any point in the sequence, but optimal in the limit, and although the conditional expectation approximation converges, the optimal stopping time will never be correctly identified. Assumption 1 ensures that this problem arises with probability zero. Thus, if $\Pr(Z(\omega, t_k) = F(\omega, t_k)) = 0$, $0 \leq k \leq K$, it is almost surely the case that if $F_M^N(\omega, t)$ converges to $F(\omega, t)$, the correct stopping time is eventually identified. In particular, at some point in the sequence, the approximation is on the same side of $Z(\omega, t_k)$ as the limit, and the approximation yields the correct stopping time.

The following proposition shows that under Assumption 1, it is, indeed, the case that the price estimate converges if the conditional expectation approximation does.

PROPOSITION 1. *Under Assumption 1, if $\hat{a}_M^N(0)$ converges to $a(0)$ as N tends to infinity, then $V_M^N(0)$ converges to $V(0)$ in probability.*

PROOF. See the appendix. $\square$

3.1. Convergence of the Conditional Expectation Approximation

Unfortunately, only a limited number of theoretical results have been obtained regarding the properties of the LSM method. One problem, of course, is the dependence introduced by the cross-sectional regressions at previous steps. This dependence limits the use of the first proposition stated in Longstaff and Schwartz (2001), because it does not hold when the coefficients are estimated. The discounted cash flows resulting from following the rule of exercising when

$Z(\omega_n, t_k)$ is positive and greater than or equal to $\hat{F}_K^M(\omega_n, t_k)$ are not independent across paths when the coefficients are estimated and common across paths. A second problem is the fact that both N and M should tend to infinity to obtain convergence of the conditional expectation approximation, $\hat{F}_M^N(\cdot, \cdot)$, to $F(\cdot, \cdot)$. This has been neglected in previous work entirely.

To keep things simple, assume for the present that the option has two periods to expiration. Apart from $\tau(T) = \tau_M(T) = T$, because $t_2 = T$, the stopping times in (3) and (7) can be written as

$$\tau(t_1) = t_1 \times 1_{\{Z(\omega, t_1) \geq F(\omega, t_1)\}} + T \times 1_{\{Z(\omega, t_1) < F(\omega, t_1)\}} \quad \text{and} \quad (14)$$

$$\tau_M(t_1) = t_1 \times 1_{\{Z(\omega, t_1) \geq F_M(\omega, t_1)\}} + T \times 1_{\{Z(\omega, t_1) < F_M(\omega, t_1)\}}. \quad (15)$$

At time t_1 , the purpose is to approximate the true conditional expectation $F(\omega, t_1)$. With known coefficients $\{a_m\}_{m=0}^\infty$, as the number of terms, M , in (6) increases, we can approximate the expectation arbitrarily well by assumption and $F_M(\omega, t_1) \rightarrow F(\omega, t_1)$ as $M \rightarrow \infty$. This result generalizes to the multiperiod situation at each t_k .

However, as mentioned, the coefficients are generally not known and must be estimated. Assume that this estimation is performed according to the LSM method. The stopping times in (10) can be written as

$$\tau_M^N(t_1) = t_1 \times 1_{\{Z(\omega, t_1) \geq \hat{F}_M^N(\omega, t_1)\}} + T \times 1_{\{Z(\omega, t_1) < \hat{F}_M^N(\omega, t_1)\}}. \quad (16)$$

Under very general conditions, it can be shown that $\hat{F}_M^N(\omega, t_1) \rightarrow F_M(\omega, t_1)$ as $N \rightarrow \infty$. This result can also be generalized to the multiperiod setting (see Clément et al. 2002), and a simplified version of Proposition 1 shows that $V_M^N(0) \rightarrow V_M(0)$ as $N \rightarrow \infty$ too.⁴

The question remains, however, under which conditions $\hat{F}_M^N(\omega, t_1)$ converges to $F(\omega, t_1)$, and whether this can be extended to more periods. Answering the first part of the question is the aim of the next section, which also gives optimal rates for N and M in the two-period case. The result will be used in §3.1.2 to show convergence of the conditional expectation approximation in the general case, thus, answering the second part of the question.

3.1.1. A Two-Period Economy. At time t_1 , we have at our disposal a sample of N observations, where suppressing t_1 for notational simplicity, $(y(\omega_n), x(\omega_n))$, $1 \leq n \leq N$, corresponds to the discounted terminal payoffs, $C(\omega_n, T)$, and the present

⁴ For approximate value functions, Theorem 2 of Tsitsiklis and Van Roy (2001) corresponds to what can be found in §3 of Clément et al. (2002) on the LSM method, because the dimension of the feature vector, which corresponds to the number of regressors M in the notation of this paper, is assumed constant throughout.

state variables along the i th path. Denote by y and x the vectors of observations. Our aim is to approximate $F(\omega)$ by

$$\hat{F}_M^N(\omega) = p_M(x)' \hat{a}_M^N, \quad (17)$$

where $p_M(x) = (p_{0,M}(x), \dots, p_{M-1,M}(x))'$ is a vector of approximating functions which has the property that a linear combination can approximate $F(\omega)$, i.e., the coordinate functions correspond to the ϕ_m from (6). The estimator from (12) may be written as

$$\hat{a}_M^N = (p_M' p_M)^{-1} p_M' y,$$

where $p_M = [p_M(x(\omega_1)), \dots, p_M(x(\omega_N))]'$. Thus, $\hat{F}_M^N(\omega)$ is recognized to be a series estimator of $F(\omega)$ in the terminology of the nonparametric statistics literature (see, e.g., Pagan and Ullah 1999). As regressors, we use powers of the x 's, and in the one-dimensional case, we would have that $p_M(x) = (1, x, x^2, \dots, x^{M-1})'$. If the dimension of x is larger than one, we specify the regressors in terms of multi-indices. Thus, a power series approximation can be defined as the following.

DEFINITION 1. Let r denote the dimension of x and let $\lambda = (\lambda_1, \dots, \lambda_r)$ denote a vector of nonnegative integers with norm $|\lambda| = \sum_{j=1}^r \lambda_j$. For a sequence of distinct such vectors, $(\lambda(m))_{m=1}^\infty$, with $|\lambda(m)|$ increasing in m , a power series approximation has

$$p_{m,M}(x) = x^{\lambda(m)}, \quad (18)$$

where $x^\lambda = \prod_{j=1}^r x_j^{\lambda_j}$ (see also Newey 1997).

An obvious example of this would be options written on multiple assets for which the level of each asset would serve as a state variable. Consider the case with $r = 2$ and $M^* = 2$, where M^* is the maximum order. The regressors used are $p_M(x) = (1, x_1, x_2, x_1^2, x_1 x_2, x_2^2)'$, i.e., the complete set of polynomials of total degree M^* in r dimensions (see Judd 1998).

For the proof of the theorem to follow, we will need the following definition and a lemma well known from the literature on series estimators.

DEFINITION 2. Let $\|A\| = [\text{trace}(A'A)]^{1/2}$ be the Euclidean norm of a matrix A , and let $|h|_0 = \sup_{x \in \mathcal{X}} |h(x)|$, where $\mathcal{X}$ denotes the support of x .

LEMMA 1 (THEOREM 1 IN NEWAY 1997). For a power series approximation, assume that the following is satisfied.

ASSUMPTION A. $(y(\omega_1), x(\omega_1)), \dots, (y(\omega_N), x(\omega_N))$ are i.i.d. and $\text{Var}(y | x)$ is bounded.

ASSUMPTION B. For every M , there is a nonsingular constant matrix B_M such that for $P_M(x) = B_M p_M(x)$: (i) the smallest eigenvalue of $E[P_M(x)P_M(x)']$ is bounded away from zero uniformly in M , and (ii) there is a sequence of constants $\zeta_0(M)$ satisfying $\sup_{x \in \mathcal{X}} \|P_M(x)\| \leq \zeta_0(M)$ and $M = M(N)$ increasing in N such that $\zeta_0(M)^2 M/N \rightarrow 0$ as $N \rightarrow \infty$.

ASSUMPTION C. There are α and a_M such that

$$|F - p_M' a_M|_0 = O(M^{-\alpha}) \text{ as } M \rightarrow \infty.$$

Then, it follows that

$$\int [F(\omega) - \hat{F}_M^N(\omega)]^2 dF_0(x) = O_p(M/N + M^{-2\alpha}),$$

where $F_0(x)$ denotes the cumulative distribution function of x .

PROOF. See Newey (1997). $\square$

REMARK 1. Assumption B imposes a normalization on the approximating functions: (i) bounds the second moment matrix away from singularity, and (ii) restricts the magnitude of the series terms and $M(N)$. Assumption C is a requirement that the approximation error shrinks at rate $M^{-\alpha}$ uniformly. The value of α is related to the smoothness of F and the dimensionality of x . In particular, for power series, one can set $\alpha = s/r$, where r is the dimension of x and s is the order of differentiability of $F(\omega)$ (see Newey 1997). The notation $O_p(M/N + M^{-2\alpha})$ conveys the notion that the ratio of the left-hand side and $M/N + M^{-2\alpha}$ remains bounded in the limit as M and N tend to infinity (see also Davidson 2000).

We can now state two assumptions under which the estimator in the cross-sectional regression in (17) can be shown to converge to the true conditional expectation function $F(\omega)$. These correspond to Assumptions 8 and 9 of Newey (1997) and are

ASSUMPTION 2. The support of x , denoted $\mathcal{X}$, is a Cartesian product of compact connected intervals on which x has a probability density function, denoted $f(x)$, that is bounded below by some $\varepsilon > 0$.

ASSUMPTION 3. The conditional expectation function is continuously differentiable of order s on the support of x .

With respect to Assumption 2, consider the simple Black-Scholes setting, where the only state variable of interest is the stock price, S . The density function is the lognormal

$$g(S) = \frac{1}{S\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\ln S - \mu)^2}{2\sigma^2}\right\},$$

where the actual values of μ and σ are unimportant for the following. When it comes to valuing the American put option, Longstaff and Schwartz (2001) suggest normalizing the stock price by the strike price and discarding all $S \geq 1$, although the density here is positive. Their motivation is that they want to approximate the conditional expectation only for the in-the-money paths, and their argument is that two to three times as many regressors are needed to obtain the same level of accuracy if all paths are used in the cross-sectional regressions. For practical purposes,

Assumption 2 is met by discarding the draws in the other tail too, $S \leq \underline{\delta}$, for a suitable $\underline{\delta} \in (0, 1)$. Thus, we define our new variable as $x = (S - \underline{\delta}) / (1 - \underline{\delta}) \in (0, 1)$ with $f(x) = (1 - \underline{\delta})g(\underline{\delta} + (1 - \underline{\delta})x) \geq \varepsilon$ for all $x \in (0, 1)$.⁵ More generally, some of the out-of-the-money paths can be included too, as long as we choose a $\underline{\delta} \geq 1$ such that draws with $S \geq \underline{\delta}$ are discarded. This poses no problem as we can choose $\underline{\delta}$ arbitrarily small and $\bar{\delta}$ arbitrarily large.⁶ Assumption 3 indicates that the smoothness of the payoff function is important.

The proof to follow uses orthonormal polynomials. After normalization, the support of the stock level, x , is in $(0, 1)$ and for this reason we choose to work with shifted Legendre polynomials, $\bar{P}_m(x)$ (see Abramowitz and Stegun 1970). Apart from $\bar{P}_0(x) = 1$, the first few shifted Legendre polynomials are $\bar{P}_1(x) = 2x - 1$, $\bar{P}_2(x) = 6x^2 - 6x + 1$, and $\bar{P}_3(x) = 20x^3 - 30x^2 + 12x - 1$. Like many other polynomial families, they satisfy a recursive formula that, in this particular case, is given by

$$\bar{P}_{m+1}(x) = \frac{(2m+1)(2x-1)}{m+1} \bar{P}_m(x) - \frac{m}{m+1} \bar{P}_{m-1}(x). \quad (19)$$

Figure 1 shows plots of the first few Legendre polynomials on the interval $(0, 1)$.

The family of shifted Legendre polynomials is orthogonal on the interval $(0, 1)$ with respect to the weighting function $w(x) = 1$; that is,

$$\int_0^1 \bar{P}_m(x) \bar{P}_l(x) dx = \begin{cases} \frac{1}{2m+1} & \text{for } m=l, \\ 0 & \text{otherwise.} \end{cases} \quad (20)$$

We normalize the polynomials by multiplying $\bar{P}_m(x)$ by $c(m) = (2m+1)^{1/2}$, and write $\tilde{P}_m(x)$ for the resulting orthonormal family.

We are now ready to state and prove our main theorem for convergence of the estimator in the cross-sectional regression in (17) to the true conditional expectation function $F(\omega)$.

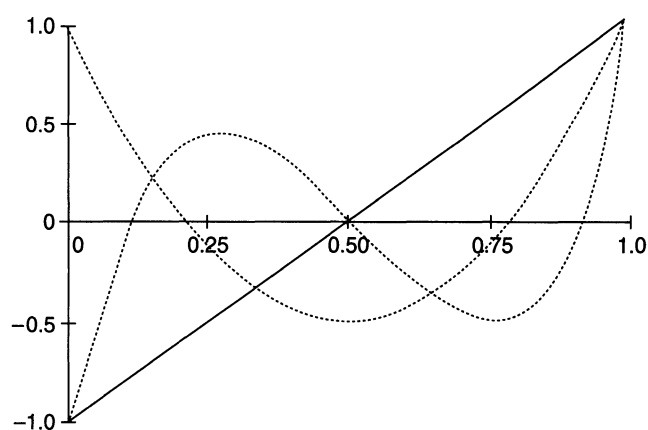
THEOREM 1. *Under Assumptions 1, 2, and 3, if $M = M(N)$ is increasing in N such that $M \rightarrow \infty$ and $M^3/N \rightarrow 0$, then the power series estimator $\hat{F}_M^N(\omega)$ in the cross-sectional regression in (17) is mean square convergent,*

$$\int [F(\omega) - \hat{F}_M^N(\omega)]^2 dF_0(x) = O_p(M/N + M^{-2s/r}), \quad (21)$$

⁵ Note that $\varepsilon = (1 - \underline{\delta}) \exp(-(\ln \underline{\delta} - \mu)^2 / (2\sigma^2)) / (\underline{\delta} \sigma \sqrt{2\pi})$ works for the put option. Note also that the same could be done for a call option by choosing some $\bar{\delta} > 1$ and setting $x = (S - 1) / (\bar{\delta} - 1)$.

⁶ Alternatively, all paths can be included by replacing Assumption 2 by the weaker Assumption B(i) from Lemma 1, which is equivalent to assumptions made in Clément et al. (2002) and Tsitsiklis and Van Roy (2001). However, in practice, setting $\underline{\delta}$ equal to the smallest tick size should circumvent any problem.

Figure 1 Plot of Shifted Legendre Polynomials $\bar{P}_m(x)$ for $m = 1, 2, 3$



where $F_0(x)$ denotes the cumulative distribution function of x , s is the order of differentiability of the conditional expectation function, and r is the dimension of x .

PROOF. See the appendix. $\square$

REMARK 2. The requirement that $M^3/N \rightarrow 0$ is what ensures the nonsingularity of the second moment matrix for the normalized shifted Legendre polynomials as N and M tend to infinity. The reason is that for this family of polynomials $\zeta_0(M) = M$ in Assumption B of Lemma 1. This rate seems to be close to the optimal one in terms of how fast M is allowed to increase. Pagan and Ullah (1999) note that the optimal rates in nonparametrics often are found to be between M^3 and M^5 . Note that the theorem does not apply if weighted Laguerre polynomials are used as regressors as in Longstaff and Schwartz (2001).

3.1.2. Convergence of the Conditional Expectation Approximation—The General Case. Clément et al. (2002) extend the convergence of $\hat{F}_M^N$ to F_M for fixed M in the two-period setting to the case of more periods by induction on t . Below, we state and prove a general convergence theorem for the conditional expectation approximation in the LSM algorithm, showing that the same can be done with Theorem 1, where M tends to infinity along with N .

THEOREM 2. *Under Assumptions 1, 2, and 3, if $M = M(N)$ is increasing in N such that $M \rightarrow \infty$ and $M^3/N \rightarrow 0$, then $\hat{F}_M^N(\omega, t_k)$ converges to $F(\omega, t_k)$ in probability for $k = 1, \dots, K$.*

PROOF. See the appendix. $\square$

Although Theorem 1 showed convergence in a mean squared sense of $\hat{F}_M^N(\omega, t_{K-1})$ to $F(\omega, t_{K-1})$, Theorem 2 deals with convergence in probability. The mean squared convergence is difficult to prove because of the pathwise dependence between the future payoffs. It might be possible to prove mean squared convergence using the framework of White and Wooldridge (1991). An alternative solution would be to assume

that a different set of paths is used at each possible time of exercise, taking the future conditional expectation functions as given. This would allow us to use Theorem 1 at each possible exercise time, and we would obtain mean squared convergence as well as actual rates of convergence for the conditional expectation approximation in terms of $K * N$ instead of N . However, as the ultimate goal is to price options consistently, and because Theorem 2 suffices for this, we do not pursue either of these extensions.

3.2. Convergence of the LSM Algorithm

From Proposition 1, it follows that the price estimate converges if the conditional expectation approximation does. Thus, Theorem 1 essentially proves that the price estimate from the LSM method converges to the true price in a two-period setting as $N \rightarrow \infty$ if $M = M(N)$ is increasing in N such that $M \rightarrow \infty$ and $M^3/N \rightarrow 0$. Compared to Proposition 2 in Longstaff and Schwartz (2001), Theorem 1 provides a significant improvement for at least three reasons. First, Theorem 1 emphasizes that both N and M should tend to infinity to obtain convergence. This is not clear from Longstaff and Schwartz (2001), and we find their statement that “the LSM algorithm converges to any desired degree of accuracy” (p. 125) somewhat problematic, because the choice of M depends crucially on the degree of accuracy, ε . Second, the theorem is applicable for arbitrary dimensions of x and immediately shows that the LSM method can be used to price options on multiple underlying assets. However, the theorem is much more general than this and can be used to show convergence of the LSM method in situations with multiple stochastic factors besides the actual asset price. Thus, Theorem 1 provides the mathematical foundation for using the LSM method to price options in models with stochastic volatility, stochastic dividends, stochastic interest rates, or even a variety of options with path-dependent payoff functions such as Asian options. These situations are not covered by Proposition 2 in Longstaff and Schwartz (2001), which can be used only in the simple Black-Scholes model. Third, with regard to the conditional expectation approximation, the theorem provides us with a rate of convergence. From this rate, an optimal relation between M and N can be derived. However, the rate of convergence depends not only on M and N , but also on the smoothness of the conditional expectation function in relation to the dimension of x . Thus, the theorem underscores the importance of smoothness of the conditional expectation function. If this function is not sufficiently smooth, the LSM estimate might not converge to the true price. We return to this in the next section.

Theorem 2, in combination with Proposition 1, proves the convergence of the LSM method in a general multiperiod setting. We note that such a result

was not obtained by Longstaff and Schwartz (2001), nor has it been possible to find any proofs of consistency when searching the literature. The papers by Clément et al. (2002) and Tsitsiklis and Van Roy (2001) neglect the fact that it is crucial to obtain consistency at this level of generality that both the number of regressors and the number of paths tend to infinity. Thus, Theorem 2 is the first to give a general proof of consistency of the LSM method.

3.2.1. Rates of Convergence. As mentioned, Theorem 1 provides us with rates of convergence for the conditional expectation approximation in the two-period situation, depending not only on the number of paths, N , and the number of regressors, M , but also on the order of differentiability of the conditional expectation function, s , in relation to r , the dimension of x . As an example, consider choosing $M \propto N^{1/4}$, which satisfies the requirement in the theorem that $M^3/N \rightarrow 0$. Theorem 1 then reads

$$\int [F(\omega) - \hat{F}_M^N(\omega)]^2 dF_0(x) = O_p(N^{-3/4} + N^{-s/2r}).$$

From this, we see that as long as the conditional expectation function is differentiable such that $s > 0$, the power series approximation converges. In the situation with one state variable, if the conditional expectation is (at least) twice continuously differentiable, which is the case for, say, a simple American-style option, the first term dominates the second one and the rate of convergence will be $N^{-3/4}$. More generally, this will be the case as long as $s > 3r/2$.

Theorem 1 does not only allow us to derive actual rates of convergence of the conditional expectation function. In addition, the theorem provides information on how to optimally choose the relation between M and N . To see this, note that the highest rate of convergence is achieved when the two terms in (21) go to zero at the same rate. Equating these implies that M should tend to infinity at the same rate as $N^{r/(r+2s)}$, again provided that $M^3/N \rightarrow 0$. Thus, the higher the order of differentiability of the conditional expectation function, the slower the number of regressors should be increased in relation to the number of paths. In a real-world situation where computational resources are scarce, this is of great importance, because in terms of mean squared error, it is worth much more to increase the number of paths than to increase the number of regressors. To get the intuition behind this, observe that the first term in the mean squared Equation (21) is related to the variance, whereas the second term is related to the bias of the approximation (see Newey 1997). For smooth functions, the bias is very small and to lower the mean squared error, the number of paths should be increased rapidly because this lowers the variance. We note that as s tends to infinity, the highest possible rate of convergence tends

to N^{-1} . As the other extreme case, consider the situation where the order of differentiability is equal to the dimension of x . This, together with the restriction that $M^3/N \rightarrow 0$, implies an optimal rate when $M \propto N^{1/(3+\gamma)}$, $\gamma > 0$. With this choice, the rate of convergence of the conditional expectation approximation can be made arbitrarily close to $N^{-2/3}$.

3.2.2. Rates of Convergence for the Price Estimates. When it comes to the price estimates, it has not been possible to derive actual rates of convergence. The problem is much the same as with Theorem 2, and is caused by the dependence between the payoff paths introduced by the cross-sectional regressions. However, because the conditional expectation approximation converges to the true function, the paths should be asymptotically independent, and it should be possible to derive a limiting distribution of the normal type.

There is a way to implement the LSM method such that rates of convergence are easily obtained, however. The strategy is to use one set of paths to calculate the conditional expectation approximation and a second set of paths to value the option. Denote the price estimated from the modified LSM method by $V_M^{N,N_2}(0)$, where N_2 denotes the number of paths used to calculate the value of the option. If the paths used for pricing are independent and identically distributed, this would imply independence and identical distribution of the pathwise payoffs. This allows us to invoke a Law of Large Numbers, and because the conditional expectation approximation converges, $V_M^{N,N_2}(0)$ converges to $V(0)$. Another benefit from this way of implementation is that we can easily invoke a central limit theorem to prove asymptotic normality of the estimator. We state this as a proposition, the proof of which should not be necessary.

PROPOSITION 2. *Under the assumptions above, we have that*

$$\sqrt{N_2}(V_M^{N,N_2}(0) - V(0)) \rightarrow N(0, \widehat{\text{Var}}(V_M^{N,N_2}(0))),$$

where $V_M^{N,N_2}(0)$ is the price estimate from the modified LSM method. The estimated variance can be calculated using the standard formula

$$\widehat{\text{Var}}(V_M^{N,N_2}(0)) = \frac{1}{N_2} \sum_{n=1}^{N_2} (C(\omega_n, 0) - \overline{C(\omega_n, 0)})^2,$$

where it is understood that $C(\omega_n, 0)$ is the time zero payoff from following the early exercise strategy derived from the coefficients $\hat{a}_M^N$.

4. Conclusion

In this paper, the asymptotic properties of the LSM method suggested in Longstaff and Schwartz (2001)

are analyzed. We prove mean squared convergence of the conditional expectation approximation to the true conditional expectation in the two-period situation and give rates of convergence. In relation to previous work, the contribution is threefold. First, the theorem emphasizes the importance of both M and N tending to infinity to obtain convergence. Second, the theorem is applicable for an arbitrary number of stochastic factors. Third, we give optimal rates for how to choose the number of regressors, M , in relation to the number of paths, N , and the optimal rates are related to the smoothness of the conditional expectation function and the number of state variables.

This paper also proves convergence of the conditional expectation approximation in the general multiperiod setting. Finally, we prove that the convergence of the conditional expectation approximation to the true expectation function results in convergence of the price estimates from the LSM method. The assumptions needed for the theorems are general. Thus, the paper provides the mathematical foundation for the use of the LSM method.

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Appendix. Proofs

A.1. Preliminaries

With the notation of §2, the following lemma, adapted from Clément et al. (2002), enables us to bound the magnitude of the difference in payoff from following different optimal stopping time strategies. The lemma will help us prove convergence of the price estimate when the conditional expectation approximation converges and help us prove convergence of the conditional expectation function in the general multiperiod setting.

LEMMA 2. *For $t_k, k = 1, \dots, K$, we have that*

$$|C(\omega, \alpha(t_k)) - C(\omega, \beta(t_k))| \leq \sum_{i=k}^K |Z(\omega, t_i)| \sum_{i=k}^{K-1} 1_{\{|Z(\omega, t_i) - \phi(X(\omega, t_i))' \beta(t_i)| \leq |\phi(X(\omega, t_i))' \beta(t_i) - \phi(X(\omega, t_i))' \alpha(t_i)|\}},$$

where α and β are vectors of coefficients and ϕ denotes a vector of transformations of $X(\omega, t_i)$.

PROOF OF LEMMA 2. Denote by $B_\alpha(t_k) = \{Z(\omega, t_k) \geq \phi(X(\omega, t_k))' \alpha(t_k)\}$ and $B_\beta(t_k) = \{Z(\omega, t_k) \geq \phi(X(\omega, t_k))' \beta(t_k)\}$. Then, ignoring any discounting terms for notational convenience, we have that

$$\begin{aligned} C(\omega, \alpha(t_k)) - C(\omega, \beta(t_k)) &= Z(\omega, t_k)(1_{B_\alpha(t_k)} - 1_{B_\beta(t_k)}) \end{aligned}$$

$$+ \sum_{i=k+1}^{K-1} Z(\omega, t_i) (1_{B_\alpha(t_k)^c \dots B_\alpha(t_{i-1})^c B_\alpha(t_i)} - 1_{B_\beta(t_k)^c \dots B_\beta(t_{i-1})^c B_\beta(t_i)}) \\ + Z(\omega, t_K) (1_{B_\alpha(t_k)^c \dots B_\alpha(t_{K-2})^c B_\alpha(t_{K-1})^c} - 1_{B_\beta(t_k)^c \dots B_\beta(t_{K-2})^c B_\beta(t_{K-1})^c}).$$

From the properties of indicator functions, we have⁷

$$|1_{B_\alpha(t_k)^c \dots B_\alpha(t_{i-1})^c B_\alpha(t_i)} - 1_{B_\beta(t_k)^c \dots B_\beta(t_{i-1})^c B_\beta(t_i)}| \\ \leq \sum_{j=k}^{i-1} |1_{B_\alpha(t_j)^c} - 1_{B_\beta(t_j)^c}| + |1_{B_\alpha(t_i)} - 1_{B_\beta(t_i)}| \\ = \sum_{j=k}^i |1_{B_\alpha(t_j)} - 1_{B_\beta(t_j)}| \quad \text{and} \\ |1_{B_\alpha(t_k)^c \dots B_\alpha(t_{K-2})^c B_\alpha(t_{K-1})^c} - 1_{B_\beta(t_k)^c \dots B_\beta(t_{K-2})^c B_\beta(t_{K-1})^c}| \\ \leq \sum_{j=k}^{K-1} |1_{B_\alpha(t_j)} - 1_{B_\beta(t_j)}|.$$

Thus, we have

$$|C(\omega, \alpha(t_k)) - C(\omega, \beta(t_k))| \\ \leq \sum_{i=k}^K |Z(\omega, t_i)| \sum_{i=k}^{K-1} |1_{B_\alpha(t_i)} - 1_{B_\beta(t_i)}|.$$

However, because $|1_A - 1_B| = 1_{(A \setminus B) \cup (B \setminus A)}$, we note that

$$|1_{B_\alpha(t_k)} - 1_{B_\beta(t_k)}| \\ = |1_{\{Z(\omega, t_k) \geq \phi(X(\omega, t_k))' \alpha(t_k)\}} - 1_{\{Z(\omega, t_k) \geq \phi(X(\omega, t_k))' \beta(t_k)\}}| \\ = 1_{\{\phi(X(\omega, t_k))' \beta(t_k) > Z(\omega, t_k) \geq \phi(X(\omega, t_k))' \alpha(t_k)\} \\ \cup \{\phi(X(\omega, t_k))' \alpha(t_k) > Z(\omega, t_k) \geq \phi(X(\omega, t_k))' \beta(t_k)\}} \\ = 1_{\{0 > Z(\omega, t_k) - \phi(X(\omega, t_k))' \beta(t_k) \geq \phi(X(\omega, t_k))' \alpha(t_k) - \phi(X(\omega, t_k))' \beta(t_k)\} \\ \cup \{\phi(X(\omega, t_k))' \alpha(t_k) - \phi(X(\omega, t_k))' \beta(t_k) > Z(\omega, t_k) - \phi(X(\omega, t_k))' \beta(t_k) \geq 0\}} \\ = 1_{\{0 < \phi(X(\omega, t_k))' \beta(t_k) - Z(\omega, t_k) \leq \phi(X(\omega, t_k))' \alpha(t_k) - \phi(X(\omega, t_k))' \beta(t_k)\} \\ \cup \{\phi(X(\omega, t_k))' \alpha(t_k) - \phi(X(\omega, t_k))' \beta(t_k) > Z(\omega, t_k) - \phi(X(\omega, t_k))' \beta(t_k) \geq 0\}} \\ = 1_{|Z(\omega, t_k) - \phi(X(\omega, t_k))' \beta(t_k)| \leq |\phi(X(\omega, t_k))' \beta(t_k) - \phi(X(\omega, t_k))' \alpha(t_k)|}.$$

Combining this with the above, we obtain Lemma 2. $\square$

Using Lemma 2, Proposition 1 is easily proved as follows.

PROOF OF PROPOSITION 1. First, observe that we have convergence if $(1/N) \sum_{n=1}^N C(\omega_n, \hat{a}_M^N(0))$ converges to $E[C(\omega, a(0))]$. However, because the simulated paths are independent by Assumption 1(i), we know that $(1/N) \cdot \sum_{n=1}^N C(\omega_n, a(0))$ converges to this expectation. Thus, it suffices to show that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N (C(\omega_n, \hat{a}_M^N(0)) - C(\omega_n, a(0))) = 0.$$

Denote by $G_N = (1/N) \sum_{n=1}^N (C(\omega_n, \hat{a}_M^N(0)) - C(\omega_n, a(0)))$. By Lemma 2, we have

$$|G_N| = \frac{1}{N} \sum_{n=1}^N |C(\omega_n, \hat{a}_M^N(0)) - C(\omega_n, a(0))| \\ \leq \frac{1}{N} \sum_{n=1}^N \sum_{i=0}^K |Z(\omega_n, t_i)| \sum_{i=0}^{K-1} 1_{\{|Z(\omega_n, t_i) - F(\omega_n, t_i)| \leq |F(\omega_n, t_i) - \hat{F}_M^N(\omega_n, t_i)|\}}.$$

⁷ In particular, recall that $1_{A^c} = 1 - 1_A$ and $1_{AB} = 1_A + 1_B - 1_A \cdot 1_B$.

However, if $\hat{a}_M^N(0)$ converges to $a(0)$ as $N \rightarrow \infty$, it must be the case that $\hat{F}_M^N(\omega, t_i) \rightarrow F(\omega, t_i)$ as $N \rightarrow \infty$. Thus, for some $\varepsilon > 0$, we get

$$\limsup_{N \rightarrow \infty} |G_N| \\ \leq \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \sum_{i=0}^K |Z(\omega_n, t_i)| \sum_{i=0}^{K-1} 1_{\{|Z(\omega_n, t_i) - F(\omega_n, t_i)| \leq \varepsilon\}} \\ = E \sum_{i=0}^K |Z(\omega, t_i)| \sum_{i=0}^{K-1} 1_{\{|Z(\omega, t_i) - F(\omega, t_i)| \leq \varepsilon\}},$$

where the last equality follows by a Law of Large Numbers given Assumption 1(i). Letting ε go to zero, we get convergence because by Assumption 1(ii), $\Pr(Z(\omega, t_k) = F(\omega, t_k)) = 0$, $0 \leq k \leq K$. $\square$

A.2. Main Theorems

We now prove the two main theorems of this paper.

PROOF OF THEOREM 1. We will prove that our framework falls within that of Lemma 1 by proving that Assumptions A, B, and C are satisfied.

(1) The observations $(y(\omega_n), x(\omega_n))$ are clearly independent and identically distributed across paths. Furthermore, the variance of the future payoff is clearly bounded after normalization. Thus, Assumption A of Lemma 1 is satisfied.

(2) It is obvious from the formula for the Legendre polynomials that if $p_M(x)$ is a power series, then there exists a nonsingular constant B such that each term in $P_M(x) = Bp_M(x)$ is the individual power replaced by the same order of the Legendre polynomial appropriately normalized. Beyond this, Assumption B has the following two parts that we deal with sequentially:

(i) From (20) and the fact that on $(0, 1)$ the density $f(x)$ is bounded away from zero, the smallest eigenvalue of $E[P_M(x)P_M(x)']$ can be calculated as

$$\lambda_{\min}(E[P_M(x)P_M(x)']) = \lambda_{\min} \left(\int_0^1 P_M(x)P_M(x)' f(x) dx \right) \\ \geq \lambda_{\min} \left(\int_0^1 P_M(x)P_M(x)' dx \times \varepsilon \right) \\ = \lambda_{\min}(I \times \varepsilon) \\ = \varepsilon > 0$$

for all M .

(ii) The constants in the Legendre polynomials were chosen to normalize them. In particular, $c(m) = (2m+1)^{1/2}$ is of order of magnitude of $m^{1/2}$. For shifted Legendre polynomials, we know that $\sup_{x \in (0,1)} |\bar{P}_m(x)| = 1$. Thus, it follows that

$$\max_{m < M} \sup_{x \in (0,1)} |P_{mM}(x)| \leq CM^{1/2}$$

and we have that $\|P_M(x)\| \leq CM = \zeta(M)$. If we choose $M = M(N)$ increasing in N such that $M^3/N \rightarrow 0$, we get that $\zeta(M)^2 M/N \rightarrow 0$ as required.

(3) Following Newey (1997) for power series, Assumption C is satisfied with $\alpha = s/r$, where r is the dimension of x and s is the order of differentiability of $F(\omega)$. $\square$

REMARK 3. It is clear from the above that the difficult part to prove is Assumption B of Lemma 1. For our proof, we need the bound on $f(x) \geq \varepsilon$. The combination of this

with the orthonormality of the Legendre polynomials with respect to the weighting function $w(x) = 1$ is what allows us to show that the minimum eigenvalue of the matrix $E[P_M(x)P_M(x)']$ is bounded away from zero.

PROOF OF THEOREM 2. Under the assumption on the conditional expectation function, this is equivalent to proving that $p_M(X(\cdot))'\hat{a}_M^N(\cdot)$ converges to $p(X(\cdot))'a(\cdot)$. This is obviously true at time $t_K = T$ and by Theorem 1, we have that $p_M(X(\omega, t_{K-1}))'\hat{a}_M^N(t_{K-1}, \hat{a}_M^N(t_K)) \rightarrow p(X(\omega, t_{K-1}))'a(t_{K-1})$. We proceed by induction. Assume that convergence holds for $i = k, \dots, K-1$. We want to show that it holds for $k-1$; that is, we have to show that $p_M(X(\omega, t_{k-1}))'\hat{a}_M^N(t_{k-1}, \hat{a}_M^N(t_k)) \rightarrow p(X(\omega, t_{k-1}))'a(t_{k-1})$. However, if we define

$$\tilde{a}_M^N(t_{k-1}, a(t_k)) = (P'P)^{-1}P'C(\omega, a(t_k)),$$

then Theorem 1 shows that $p_M(X(\omega, t_{k-1}))'\hat{a}_M^N(t_{k-1}, a(t_k)) \rightarrow p(X(\omega, t_{k-1}))'a(t_{k-1})$ as $N \rightarrow \infty$. Thus, comparing $\hat{a}_M^N(t_{k-1}, a(t_k))$ to $\hat{a}_M^N(t_{k-1}, \hat{a}_M^N(t_k))$, we see that it suffices to show that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N (p_M(X(\omega_n, t_{k-1}))'C(\omega_n, \hat{a}_M^N(t_k, \hat{a}_M^N(t_{k+1}))) - p_M(X(\omega_n, t_{k-1}))'C(\omega_n, a(t_k))) = 0.$$

Denote by

$$G_N = \frac{1}{N} \sum_{n=1}^N (p_M(X(\omega_n, t_{k-1}))'C(\omega_n, \hat{a}_M^N(t_k, \hat{a}_M^N(t_{k+1}))) - p_M(X(\omega_n, t_{k-1}))'C(\omega_n, a(t_k))).$$

By Lemma 2, we have that

$$\begin{aligned} |G_N| &= \frac{1}{N} \sum_{n=1}^N |p_M(X(\omega_n, t_{k-1}))'C(\omega_n, \hat{a}_M^N(t_k, \hat{a}_M^N(t_{k+1}))) - p_M(X(\omega_n, t_{k-1}))'C(\omega_n, a(t_k))| \\ &\leq \frac{1}{N} \sum_{n=1}^N |p_M(X(\omega_n, t_{k-1}))| \\ &\quad \cdot |C(\omega_n, \hat{a}_M^N(t_k, \hat{a}_M^N(t_{k+1}))) - C(\omega_n, a(t_k))| \\ &\leq \frac{1}{N} \sum_{n=1}^N |p_M(X(\omega_n, t_{k-1}))| \sum_{i=k}^K |Z(\omega_n, t_i)| \\ &\quad \cdot \sum_{i=k}^{K-1} 1_{\{|Z(\omega_n, t_i) - p(X(\omega_n, t_i))'a(t_i)| \leq |p(X(\omega_n, t_i))'a(t_i) - p_M(X(\omega_n, t_i))'\hat{a}_M^N(t_i, \hat{a}_M^N(t_{i+1}))|\}} \end{aligned}$$

Because by assumption $p_M(X(\omega, t_i))'\hat{a}_M^N(t_i, \hat{a}_M^N(t_{i+1}))$ converges to $p(X(\omega, t_i))'a(t_i)$ for $i = k, \dots, K-1$, we get for some $\varepsilon > 0$,

$$\begin{aligned} \limsup_{N \rightarrow \infty} |G_N| &\leq \limsup_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N |p_M(X(\omega_n, t_{k-1}))| \\ &\quad \cdot \sum_{i=k}^K |Z(\omega_n, t_i)| \sum_{i=k}^{K-1} 1_{\{|Z(\omega_n, t_i) - p(X(\omega_n, t_i))'a(t_i)| \leq \varepsilon\}} \\ &= E|p_M(X(\omega, t_{k-1}))| \\ &\quad \cdot \sum_{i=k}^K |Z(\omega, t_i)| \sum_{i=k}^{K-1} 1_{\{|Z(\omega, t_i) - p(X(\omega, t_i))'a(t_i)| \leq \varepsilon\}}, \end{aligned}$$

where the last equality follows from a Law of Large Numbers since the stock price and, hence, also the projections of the stock prices, and payoffs are independent and identically distributed along each path by Assumption 1(i). Letting ε go to zero, we get convergence because by Assumption 1(ii), $\Pr(Z(\omega, t_k) = p(X(\omega, t_k))'a(t_k)) = 0$. $\square$

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